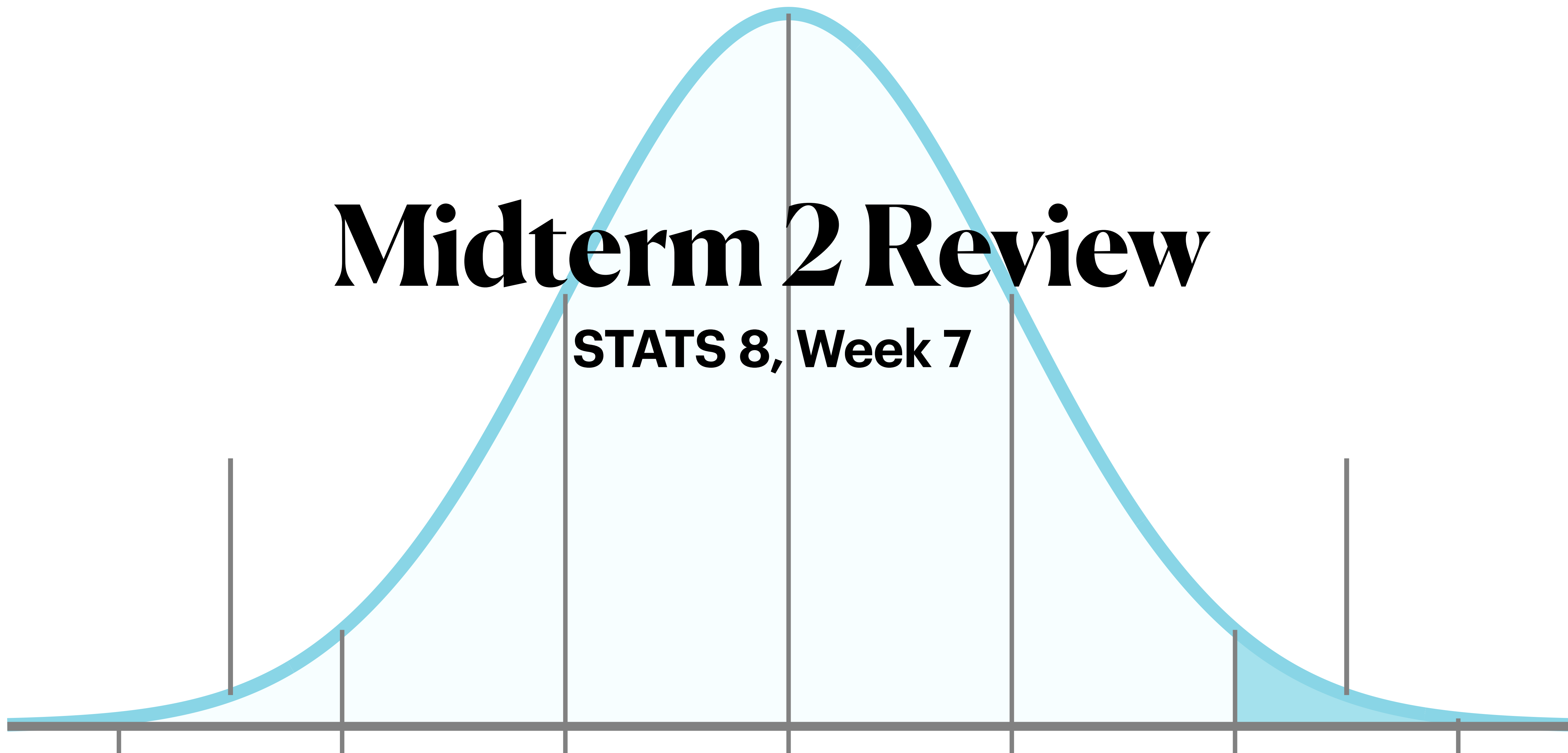


Midterm 2 Review

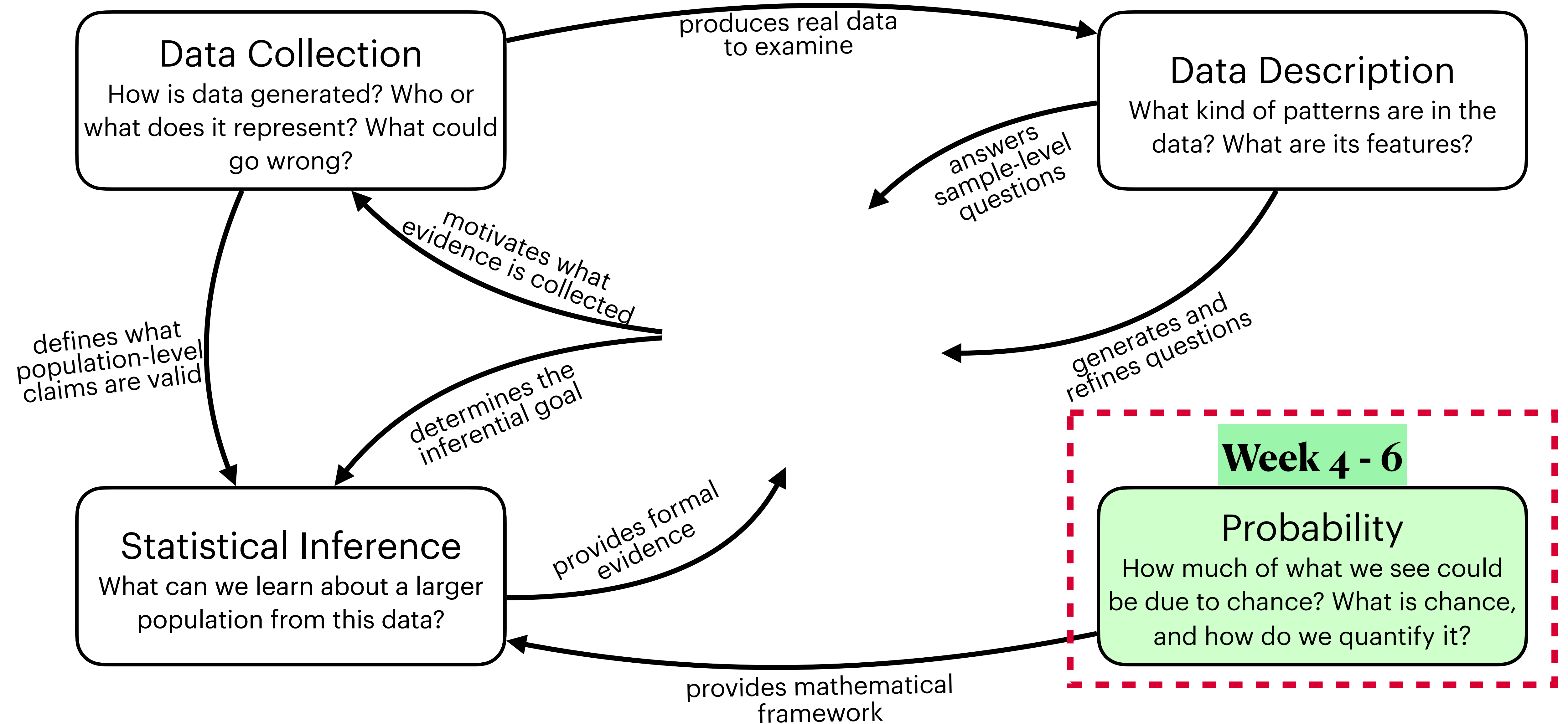
STATS 8, Week 7



How are you feeling about this midterm?

[PollEv.com/stats8](https://poll-ev.com/stats8)

- A. Great, could do it all with my eyes closed
- B. Pretty good, I think I will do fairly well
- C. Okay, I think I understand stuff but I need to do a lot of studying
- D. Not great, I think that even if I continue practicing I won't do very well
- E. Pretty bad, I'm struggling to follow and I don't think I'll be able to pass
- F. Absolutely abysmal, I'm completely lost and don't know what to do



Topics to Know

Topic 4: Probability Concepts

Identifying the random circumstance and sample space

Notation: $P(A)$, $P(A^c)$, $P(A \text{ or } B)$, $P(A \text{ and } B)$, $P(A \mid B)$

Understanding and using complements, unions, intersections, addition rule, law of total probability, Venn diagrams

Recognizing mutually exclusive events and independent events

Topic 5: Probability Dists + RVs

Understanding what makes a quantity random

Using a probability distribution table/pdf and cdf

Computing expected value and variance for a discrete random variable

Recognizing and working with Bernoulli, Uniform, and Normal distributions

Topic 6: Sampling Dist + CLT

Distinguishing individual values from sample means/proportions

Finding the center and standard deviation of a sampling distribution for a mean and proportion

Explaining why sample means vary and why larger n reduces variability

Knowing what the CLT says and what it does not say

Practice Questions

+ PollEverywhere

How this review will go:

1. Answer the questions on the slide.
 2. Go to PollEverywhere and answer the extra question once you are done, so I can see when people are wrapping up.
 3. We go over the solutions.
 4. We move on to the next set of questions, and repeat.
- ✱ We will need to move fast to get through as much as we can in a short lecture! We may not get through all of it, but answers are given at the end of these slides, and if anything is confusing — come to office hours/email me.

Probability Concepts

1. Set Theory with Dice (3 min)

- A standard, fair six-sided die is rolled once. Let A be the event that the roll is odd, and B be the event that the roll is greater than 4.
 - A. Write the sample space as a set.
 - B. Write A and B as sets.
 - C. Write (A or B) and (A and B) as sets.
 - D. What is $P(A \text{ and } B)$?
 - E. Are event A and event B independent?
 - F. Are event A and event B mutually exclusive?

Probability Concepts

PollEv Check 1

- What is the difference between independent and mutually exclusive events?
 - A. Mutually exclusive means the events cannot both happen; independent means knowing one event happened does not change the probability of the other.
 - B. Independent means the events cannot both happen; mutually exclusive means knowing one event happened does not change the probability of the other.
 - C. Mutually exclusive events are events with no overlap, while independent events are events with no probability.
 - D. Independent and mutually exclusive mean basically the same thing: the events do not affect each other.

Probability Concepts

PollEv Check 1

- What is the difference between independent and mutually exclusive events?
 - A. **Mutually exclusive means the events cannot both happen; independent means knowing one event happened does not change the probability of the other.**
 - B. Independent means the events cannot both happen; mutually exclusive means knowing one event happened does not change the probability of the other.
 - C. Mutually exclusive events are events with no overlap, while independent events are events with no probability.
 - D. Independent and mutually exclusive mean basically the same thing: the events do not affect each other.

Probability Concepts

2. Caffeine (6 mins)

- In a survey, **38%** of students say they drink **coffee**, **24%** say they drink **energy drinks**, and **10%** say they drink **both**. Suppose I randomly select one student from this survey. Let **C** be the event that they drink coffee, and let **E** be the event that they drink energy drinks.
 - A. What is $P(E)$?
 - B. What is $P(C \text{ and } E)$?
 - C. What is $P(C \text{ or } E)$?
 - D. Are event C and event E independent?
 - E. Are event C and event E mutually exclusive?
 - F. What is $P(E \mid C)$?
 - G. What does $P(E^c \text{ and } C^c)$ represent, in plain English?

Probability Concepts

PollEv Check 2

- What is the point of “conditional probability”?
 - A. It tells us the probability that two events happen at the exact same time.
 - B. It tells us whether two events are mutually exclusive.
 - C. It tells us the probability of an event after removing all outcomes where that event does not happen.
 - D. It tells us the probability of one event, given that we know another event has happened.
 - E. It tells us the probability of one event assuming the two events are independent.

Probability Concepts

PollEv Check 2

- What is the point of “conditional probability”?
 - A. It tells us the probability that two events happen at the exact same time.
 - B. It tells us whether two events are mutually exclusive.
 - C. It tells us the probability of an event after removing all outcomes where that event does not happen.
 - D. It tells us the probability of one event, given that we know another event has happened.**
 - E. It tells us the probability of one event assuming the two events are independent.

Probability Distributions and Random Variables

3. Number of Pets in a Household (pdf) (4 min)

- A researcher plans to randomly select one household in a large neighborhood. Let X = number of pets owned by the randomly selected household. Suppose the distribution is:

x	0	1	2	3
$P(X = x)$	0.35	0.30	?	0.10

- A. What makes X random?
- B. Fill in the missing probability.
- C. What is $P(X \geq 2)$?
- D. What is $P(X > 2)$?

Probability Distributions and Random Variables

PollEv Check 3

- After the household is selected, is the number of pets in the household random?
 - A. Yes, because the household was selected randomly.
 - B. Yes, because we may not know how many pets they have yet.
 - C. No, because once the household is selected, the number of pets is fixed, even if we do not know it yet.
 - D. No, because numbers of pets are discrete, not random.

Probability Distributions and Random Variables

PollEv Check 3

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 - A. Yes, because the household was selected randomly.
 - B. Yes, because we may not know how many pets they have yet.
 - C. No, because once the household is selected, the number of pets is fixed, even if we do not know it yet.**
 - D. No, because numbers of pets are discrete, not random.

Probability Distributions and Random Variables

4. Number of Pets in a Household (cdf) (4 min)

- A researcher plans to randomly select one household in a large neighborhood. Let X = number of pets owned by the randomly selected household. Suppose the distribution is:

x	0	1	2	3
$P(X \leq x)$				

- A. Given the pdf from the previous slides, fill in the cdf of X .
- B. Which cell in the cdf do I look at if I want $P(X \leq 2)$?
- C. Which cell in the cdf do I look at if I want $P(X < 2)$?
- D. Which cells in the cdf should I subtract if I want $P(1 \leq X \leq 2)$?

Probability Distributions and Random Variables

PollEv Check 4

- Why is the last value in a cdf always 1?
 - A. Because the last value includes all possible values of the random variable, so the probability is 1.
 - B. Because the largest possible value of any random variable always has probability 1.
 - C. Because the probabilities in a cdf must all be the same size.
 - D. Because the cdf gives the probability that the random variable equals each value, and one of the values must happen.
 - E. Because cdfs are always increasing, so the last value has to be 1.

Probability Distributions and Random Variables

PollEv Check 4

- Why is the last value in a cdf always 1?
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 - B. Because the largest possible value of any random variable always has probability 1.
 - C. Because the probabilities in a cdf must all be the same size.
 - D. Because the cdf gives the probability that the random variable equals each value, and one of the values must happen.
 - E. Because cdfs are always increasing, so the last value has to be 1.

Probability Distributions and Random Variables

5. Number of Pets in a Household ($E()$ and $\text{Var}()$) (4 min)

- A researcher plans to randomly select one household in a large neighborhood. Let X = number of pets owned by the randomly selected household. Suppose the distribution is:

x	0	1	2	3
$P(X = x)$	0.35	0.30	0.25	0.10

- A. What is $E(X)$?
- B. What does $E(X)$ represent?
- C. What is $\text{Var}(X)$?

Probability Distributions and Random Variables

PollEv Check 5

- What is the standard deviation of X ? How is it related to the variance?
 - A. The standard deviation is 0.99; it is the same thing as the variance.
 - B. The standard deviation is about 0.995; it is the square root of the variance.
 - C. The standard deviation is about 0.980; it is the variance squared.
 - D. The standard deviation is 1.10; it is the expected value of X .
 - E. The standard deviation cannot be found from the probability distribution table.

Probability Distributions and Random Variables

PollEv Check 5

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 - D. The standard deviation is 1.10; it is the expected value of X .
 - E. The standard deviation cannot be found from the probability distribution table.

Probability Distributions and Random Variables

6. Commuters to UCI (3 mins)

- In a class of 200 students, 72 report that they commute to campus. One student is randomly selected. Let $X = 1$ if the selected student commutes, and $X = 0$ otherwise.
 - Is X Bernoulli-distributed? Why?
 - What is $P(X = 0)$?
 - What are $E(X)$ and $\text{Var}(X)$?

Probability Distributions and Random Variables

PollEv Check 6

- What makes a random variable Bernoulli?
 - A. It can take exactly two values, coded 1 and 0, with probabilities p and $1 - p$.
 - B. It is a proportion with probability p .
 - C. It can take any value between 0 and 1.
 - D. It has a probability distribution where all outcomes are equally likely.
 - E. It has a probability distribution shaped like a bell curve.
 - F. It is any random variable where the CLT applies.

Probability Distributions and Random Variables

PollEv Check 6

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 - E. It has a probability distribution shaped like a bell curve.
 - F. It is any random variable where the CLT applies.

Probability Distributions and Random Variables

7. Waiting for the Bus (5 min)

- A bus is scheduled to arrive every 12 minutes. Suppose your waiting time X is equally likely to be anywhere from 0 to 12 minutes.
 - A. What is the distribution of X ? Use the notation we learned in class.
 - B. What is $P(0 \leq X \leq 3)$?
 - C. What is $P(X \geq 10)$?
 - D. What is $P(X = 5)$?
 - E. Why is part (D) different from the discrete examples?

Probability Distributions and Random Variables

PollEv Check 7

- Why don't we use a table to describe probabilities for continuous random variables the way we do for discrete ones?
 - A. Because it's not possible to define probabilities in the context of continuous random variables.
 - B. Because continuous random variables are always Normally distributed.
 - C. Because each exact value of a continuous random variable usually has a positive probability.
 - D. Because continuous random variables can take infinitely many values, so we describe probabilities over intervals instead of listing every possible value, which is impossible.
 - E. Because tables only work when all outcomes are equally likely.

Probability Distributions and Random Variables

PollEv Check 7

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 - D. Because continuous random variables can take infinitely many values, so we describe probabilities over intervals instead of listing every possible value, which is impossible.**
 - E. Because tables only work when all outcomes are equally likely.

Sampling Distributions and CLT

8. Screen Time (3 mins)

- Suppose the mean daily screen time in a population is 5 hours with standard deviation of 2 hours. A researcher plans to randomly sample 25 people and compute the sample mean.
 - A. Is $\bar{X}$ random before the sample is selected? Why?
 - B. Around what value will the sampling distribution of $\bar{X}$ be centered?
 - C. What is the standard deviation of the sampling distribution of $\bar{X}$?
 - D. What would happen to the standard deviation of the sampling distribution of $\bar{X}$ if I increased the sample size from 25 to 144?

Sampling Distributions and CLT

PollEv Check 8

- What quality of the randomly sampled data allows us to use those rules for expected value and variance/standard deviation?
 - A. The sample size is large enough for the CLT to apply.
 - B. The sampled observations are independent, and they come from the same population distribution (i.i.d.).
 - C. The population distribution is exactly Normal.
 - D. The sample mean is close to the population mean.
 - E. The population standard deviation is known.

Sampling Distributions and CLT

PollEv Check 8

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 - E. The population standard deviation is known.

Sampling Distributions and CLT

9. Off-Campus Living (5 mins)

- Suppose 40% of students at a university live off campus. A random sample of $n = 200$ students is taken, and $\hat{P}$ is the sample proportion who live off campus.
 - A. Why is $\hat{P}$ a type of sample mean?
 - B. Do the usual CLT rule-of-thumb checks pass?
 - C. What is the approximate distribution of $\hat{P}$ in this case?
 - D. Without using any computing software, which is smaller: $P(\hat{P} \geq 0.9)$ or $P(\hat{P} \geq 0.7)$?

Sampling Distributions and CLT

PollEv Check 9

- A student says: “Because the CLT applies when the sample size n is large, the population distribution becomes approximately Normally distributed as n increases.” Do you agree or disagree?
 - A. Agree; the CLT says that large samples make the population distribution approximately Normal.
 - B. Agree; as n increases, individual observations become less variable and more Normal.
 - C. Disagree; the CLT says the sampling distribution of the sample mean or sample proportion becomes approximately Normal, not the population distribution.
 - D. Disagree; the CLT only tells us the sample mean equals the population mean when n is large.

Sampling Distributions and CLT

PollEv Check 9

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 - A. Agree; the CLT says that large samples make the population distribution approximately Normal.
 - B. Agree; as n increases, individual observations become less variable and more Normal.
 - C. Disagree; the CLT says the sampling distribution of the sample mean or sample proportion becomes approximately Normal, not the population distribution.**
 - D. Disagree; the CLT only tells us the sample mean equals the population mean when n is large.

Mixed Review

“Identify the tool”

For each task, identify which tool/equation/strategy/theorem discussed in class that you might use to approach the task.

1. Finding $P(A \text{ or } B)$ if we know $P(A)$, $P(B)$, and $P(A \text{ and } B)$.
2. Finding the long-run average payout of a gambling game with a given probability distribution of winnings per round.
3. Finding $P(4 \leq X \leq 7)$ when X is a continuous random variable equally likely between 0 and 10.
4. Finding how much sample means vary across repeated samples.
5. Explaining why $\hat{p}$ is approximately normal for large n .

Correct Answers

Probability Concepts

1. Set Theory with Dice

- A standard, fair six-sided die is rolled once. Let A be the event that the roll is odd, and B be the event that the roll is greater than 4.
 - A. Write the sample space as a set. $S = \{1, 2, 3, 4, 5, 6\}$
 - B. Write A and B as sets. $A = \{1, 3, 5\}; B = \{5, 6\}$
 - C. Write (A or B) and (A and B) as sets. $(A \text{ or } B) = \{1, 3, 5, 6\}; (A \text{ and } B) = \{5\}$
 - D. What is $P(A \text{ and } B)$? $P(A \text{ and } B) = 1/6$
 - E. Are event A and event B independent? $P(A | B) = 1/2; P(A) = 1/2$. Yes!
 - F. Are event A and event B mutually exclusive? $P(A \text{ and } B) = 1/6$. No!

Probability Concepts

2. Caffeine

- In a survey, 38% of students say they drink **coffee**, 24% say they drink **energy drinks**, and 10% say they drink **both**. Suppose I randomly select one student from this survey. Let **C** be the event that they drink coffee, and let **E** be the event that they drink energy drinks.
 - A. What is $P(E)$? **$P(E) = 0.24$**
 - B. What is $P(C \text{ and } E)$? **$P(C \text{ and } E) = 0.10$**
 - C. What is $P(C \text{ or } E)$? **$P(C \text{ or } E) = 0.38 + 0.24 - 0.10 = 0.52$**
 - D. Are event C and event E independent? **$P(C \text{ and } E) = 0.10$; $P(C)P(E) = 0.0912$. No!**
 - E. Are event C and event E mutually exclusive? **$P(C \text{ and } E) = 0.10$. No!**
 - F. What is $P(E | C)$? **$P(E | C) = P(E \text{ and } C) / P(C) = 0.2632$**
 - G. What does $P(E^c \text{ and } C^c)$ represent, in plain English? **The probability that the selected student drinks neither coffee nor energy drinks.**

Probability Distributions and Random Variables

3. Number of Pets in a Household (pdf)

- A researcher plans to randomly select one household in a large neighborhood. Let X = number of pets owned by the randomly selected household. Suppose the distribution is:

x	0	1	2	3
$P(X = x)$	0.35	0.30	0.25	0.10

- A. What makes X random? **Household is randomly sampled**
- B. Fill in the missing probability. **0.25**
- C. What is $P(X \geq 2)$? **$0.25 + 0.10 = 0.35$**
- D. What is $P(X > 2)$? **0.10**

Probability Distributions and Random Variables

4. Number of Pets in a Household (cdf)

- A researcher plans to randomly select one household in a large neighborhood. Let X = number of pets owned by the randomly selected household. Suppose the distribution is:

x	0	1	2	3
$P(X \leq x)$	0.35	0.65	0.90	1.00

- Given the pdf from the previous slides, fill in the cdf of X .
- Which cell in the cdf do I look at if I want $P(X \leq 2)$? **2nd to last column (2)**
- Which cell in the cdf do I look at if I want $P(X < 2)$? **3rd to last column (1)**
- Which cells in the cdf should I subtract if I want $P(1 \leq X \leq 2)$? **2nd to last column (2) minus 1st column (0)**

Probability Distributions and Random Variables

5. Number of Pets in a Household ($E()$ and $\text{Var}()$)

- A researcher plans to randomly select one household in a large neighborhood. Let X = number of pets owned by the randomly selected household. Suppose the distribution is:

x	0	1	2	3
$P(X = x)$	0.35	0.30	0.25	0.10

- A. What is $E(X)$? $E(X) = 0 \times P(X = 0) + \dots + 3 \times P(X = 3) = \mathbf{0.3 + 0.5 + 0.3 = 1.1}$
- B. What does $E(X)$ represent? **Avg. number of pets in many many random selections of one household.**
- C. What is $\text{Var}(X)$? $\mathbf{\text{Var}(X) = (0 - 1.1)^2 \times P(X = 0) + \dots + (3 - 1.1)^2 \times P(X = 3)}$
 $\mathbf{= 1.21(0.35) + 0.01(0.30) + 0.81(0.25) + 3.61(0.10) = 0.99}$

Probability Distributions and Random Variables

6. Commuters to UCI

- In a class of 200 students, 72 report that they commute to campus. One student is randomly selected. Let $X = 1$ if the selected student commutes, and $X = 0$ otherwise.
 - Is X Bernoulli-distributed? Why? **Yes, takes exactly 2 values (“yes” or “no”, 1 or 0) with some probability**
 - What is $P(X = 0)$? **$P(X = 0) = 1 - P(X = 1) = 1 - (72/200) = 0.64$**
 - What are $E(X)$ and $\text{Var}(X)$? **For Bernoulli, $E(X) = p$ and $\text{Var}(X) = p(1-p)$, so $E(X) = 0.36$ and $\text{Var}(X) = (0.36)(0.64) = 0.23$. (Or, you could just use the formulas for expectation and variance and get the same thing.)**

Probability Distributions and Random Variables

7. Waiting for the Bus

- A bus is scheduled to arrive every 12 minutes. Suppose your waiting time X is equally likely to be anywhere from 0 to 12 minutes.
 - A. What is the distribution of X ? Use the notation we learned in class. **$X \sim \text{Uniform}(0,12)$**
 - B. What is $P(0 \leq X \leq 3)$? **Height of pdf is $1/12$; Area of rectangle is $(1/12)(3) = 0.25$**
 - C. What is $P(X \geq 10)$? **$(1/12)(2) = 0.167$**
 - D. What is $P(X = 5)$? **0**
 - E. Why is part (D) different from the discrete examples? **Continuous random variables are different from discrete random variables: probabilities come from areas under the density curve, and the “area” under one exact value is 0.**

Sampling Distributions and CLT

8. Screen Time

- Suppose the mean daily screen time in a population is 5 hours with standard deviation of 2 hours. A researcher plans to randomly sample 25 people and compute the sample mean.
 - A. Is $\bar{X}$ random before the sample is selected? Why? **Yes, it will be the average screen time in the sample, but the sample is randomly selected so the randomness is not “resolved” yet**
 - B. Around what value will the sampling distribution of $\bar{X}$ be centered? **5**
 - C. What is the standard deviation of the sampling distribution of $\bar{X}$? **$2/5 = 0.4$**
 - D. What would happen to the standard deviation of the sampling distribution of $\bar{X}$ if I increased the sample size from 25 to 144? **At 25, the SD is $2/5 = 0.4$. At 144, it is $2/12 = 0.167$. Much smaller!**

Sampling Distributions and CLT

9. Off-Campus Living

- Suppose 40% of students at a university live off campus. A random sample of $n = 200$ students is taken, and $\hat{P}$ is the sample proportion who live off campus.
 - A. Why is $\hat{P}$ a type of sample mean? **Because it is the average of a list of 0 and 1 values, so the computation gives the proportion of 1s in the list.**
 - B. Do the usual CLT rule-of-thumb checks pass? **Yes: $0.4(200) = 80 > 10$; $0.6(200) = 120 > 10$. We also have that the sample is random from a large population, so we have i.i.d.**
 - C. What is the approximate distribution of $\hat{P}$ in this case? **$\hat{P} \sim \text{Normal}(\text{mean} = 0.40, \text{variance} = 0.0012)$**
 - D. Without using any computing software, which is smaller: $P(\hat{P} \geq 0.9)$ or $P(\hat{P} \geq 0.7)$? **The former, $P(\hat{P} \geq 0.9)$. Draw the distribution to see why!**

Mixed Review

“Identify the tool”

For each task, identify which tool/equation/strategy/theorem discussed in class that you might use to approach the task.

1. Finding $P(A \text{ or } B)$ if we know $P(A)$, $P(B)$, and $P(A \text{ and } B)$. **Addition rule or Venn diagram**
2. Finding the long-run average payout of a gambling game with a given probability distribution of winnings per round. **Expected value**
3. Finding $P(4 \leq X \leq 7)$ when X is a continuous random variable equally likely between 0 and 10. **Area under pdf of uniform distribution (area of rectangle)**
4. Finding how much sample means vary across repeated samples. **Variance or standard deviation of the sampling distribution**
5. Explaining why $\hat{p}$ is approximately normal for large n . **CLT!**